EJIE Data Space · Euskadi

A connected health system for Euskadi: what we learned from Catena-X, and what it means for the EHDS

EJIE · Basque Government · revised 4 November 2026



Your eight questions: one sentence each

The short answers first. Each one has its own slide.

01

CHALLENGES

Culture, not technology.

02

LESSONS

Build on existing standards.

03

DECISIONS

Open standards, decided in public.

04

ARCHITECTURE

Reuse components first, build what is missing.

05

MATURITY

Protocol stable, secure environments young.

06

GOVERNANCE

Keep rules and operation apart.

07

COSTS

No licences. You pay for people and hosting.

08

TEAM

8 to 15 people for a pilot.

QUESTION 1 · CHALLENGES

Culture was harder than technology

Four challenges and the source of each lesson learned.

Culture

FROM CATENA-X

Working groups delivered only when top management gave its experts time and a mandate. In a hospital the same holds for clinicians and clinical coders.

Onboarding

FROM CATENA-X

Joining was contracts, identity and a certificate check first, software second. A small participant needs a hosted connector, not its own engineers.

Governance

FROM CATENA-X

Without a rulebook nobody could say who may join, or what happens when a rule is broken. Health adds a layer Catena-X never had: the data permit.

Meaning of the data

FROM OUR EHDS DEMONSTRATOR

Mapping to FHIR, SNOMED CT, ICD-10 and OMOP took more work than the connector. Identity and keys failed more often than the data exchange.

Moving data is easy. Agreeing on what it means is hard.

QUESTION 1 · CHALLENGES · AN EXAMPLE

Example: a hospital cancer registry joins

What a hospital actually does and how its local codes become standard ones.

WHAT THE HOSPITAL DOES

1. Signs the participation agreement and the rulebook.
2. Gets its identity: an organisation credential, for example from an FNMT certificate.
3. Gets a connector: hosted by the operator, or its own.
4. Maps the registry to FHIR and OMOP, with a clinical coder.
5. Describes the dataset in the HealthDCAT-AP catalogue.
6. Passes the conformity tests, then goes live.

STEP 4: HOW LOCAL CODES ARE MAPPED

IN THE HOSPITAL	STANDARD	IN OMOP
Diagnosis, CIE-10-ES C50.912	ICD-10 C50.9 · SNOMED CT 254837009, malignant neoplasm of breast	condition
Tumour, ICD-O-3 C50.4 and 8500/3	ICD-O-3, kept as it is	oncology extension
Lab, local code "HER2 IHQ"	LOINC 18474-7, HER2 by immune stain	measurement
Stage, text "T2 N1 M0"	TNM, coded in SNOMED CT	measurement

A terminology service proposes each mapping, a clinical coder approves it, and every hospital reuses the approved table.

FROM OUR EHDS DEMONSTRATOR the same pipeline, FHIR to OMOP, on synthetic data.

QUESTION 2 · LESSONS

Four lessons to avoid common mistakes

What we would do again, where we learned it, and what we recommend for Euskadi.

1

Build on what exists

Run open working groups on FHIR, SNOMED CT, ICD-10, OMOP and HealthDCAT-AP. Do not invent new standards.

FROM CATENA-X

FROM OUR EHDS DEMONSTRATOR

2

Start small

Choose one or two use cases. Show value that you can measure.

FROM CATENA-X

3

Budget for meaning

Plan most of the effort for mapping and clinical coders, not for the connector.

FROM OUR EHDS DEMONSTRATOR

4

Buy standards, not products

Tender on standards and conformity tests. Give improvements back to the open-source projects.

FROM CATENA-X

A private copy of open source becomes your own product to maintain.

QUESTION 3 · DECISIONS

Every decision is recorded publicly

How we decided, and the five criteria behind every decision.

Working groups
propose

→

The people who build it
decide the priorities

→

A decision record
is published for everyone

FIVE CRITERIA FOR EVERY DECISION

- Open standards
- Data sovereignty
- Fits the law: EHDS, GDPR, Data Act
- Mature technology
- A clear exit strategy

Our EHDS demonstrator works the same way: more than 70 decision records, public on GitHub.

See it yourself: the demonstrator · its decision records

QUESTION 4 · ARCHITECTURE

A possible reference implementation

The components by function. Technologies are only examples; none is chosen yet.

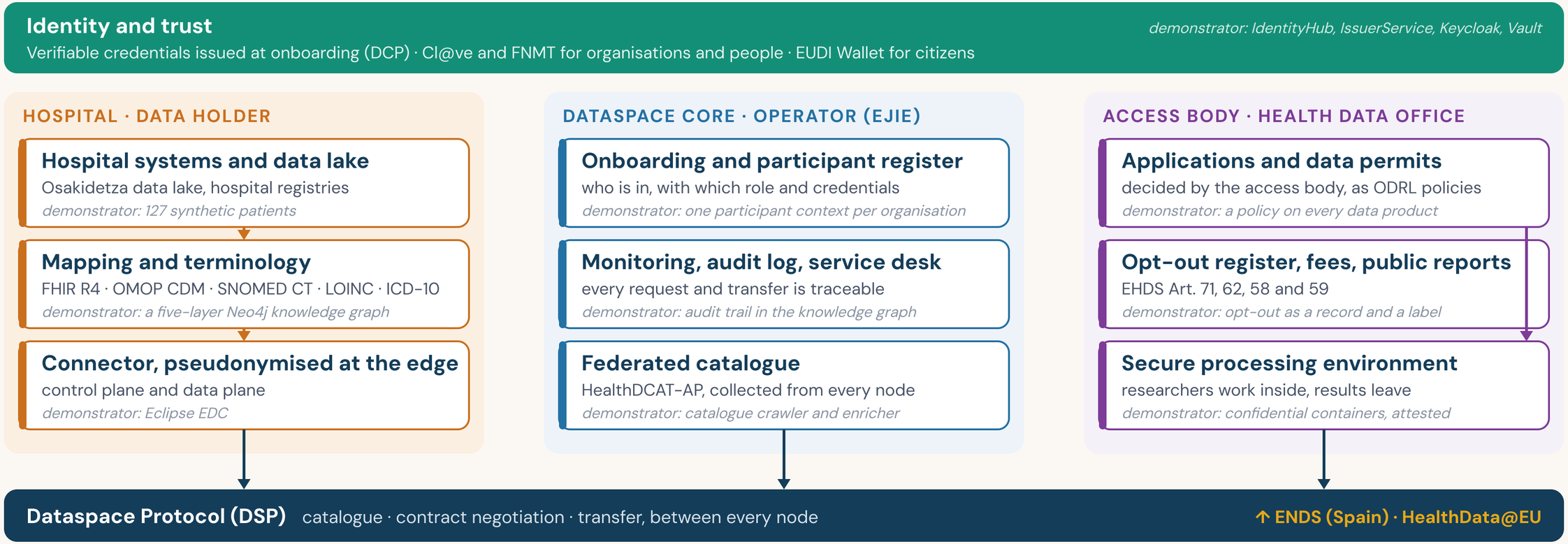
AT EACH HOSPITAL	reuse **Source systems and data lake** the Osakidetza data lake, hospital systems	adapt **Mapping to FHIR and OMOP** for example OHDSI WhiteRabbit and Usagi	adapt **Pseudonymisation** for example a trusted third-party service	reuse **Terminology service** for example a FHIR terminology server	reuse **Dataspace connector** for example Eclipse EDC or the Simpl-Open agent
SHARED, RUN BY THE OPERATOR	adapt **Identity and credentials** for example DCP credentials, linked to Cl@ve and FNMT	reuse **Dataset catalogue** HealthDCAT-AP; for example the HealthData@EU catalogue code	adapt **Onboarding and participant register** the Catena-X onboarding service as a model	adapt **Conformity tests** for example the Dataspace Protocol test kit	reuse **Monitoring and audit log** for example OpenTelemetry and a log store
FOR THE ACCESS BODY	adapt **Applications and data permits** a case workflow on the EU application form	build **Opt-out register** EHDS Art. 71: applied before any data leaves	build **Fees and public reports** EHDS Art. 62, 58 and 59	adapt **Data quality and utility label** taken from the catalogue metadata	open decision **Secure processing environment** own or ENDS; for example confidential containers

reuse use as it is · adapt configure or extend · build nothing fits today · open decision

QUESTION 4 · ARCHITECTURE

Our architecture: how the solution could be built

The EHDS demonstrator, laid out for Euskadi: three parties, one trust layer, one protocol.

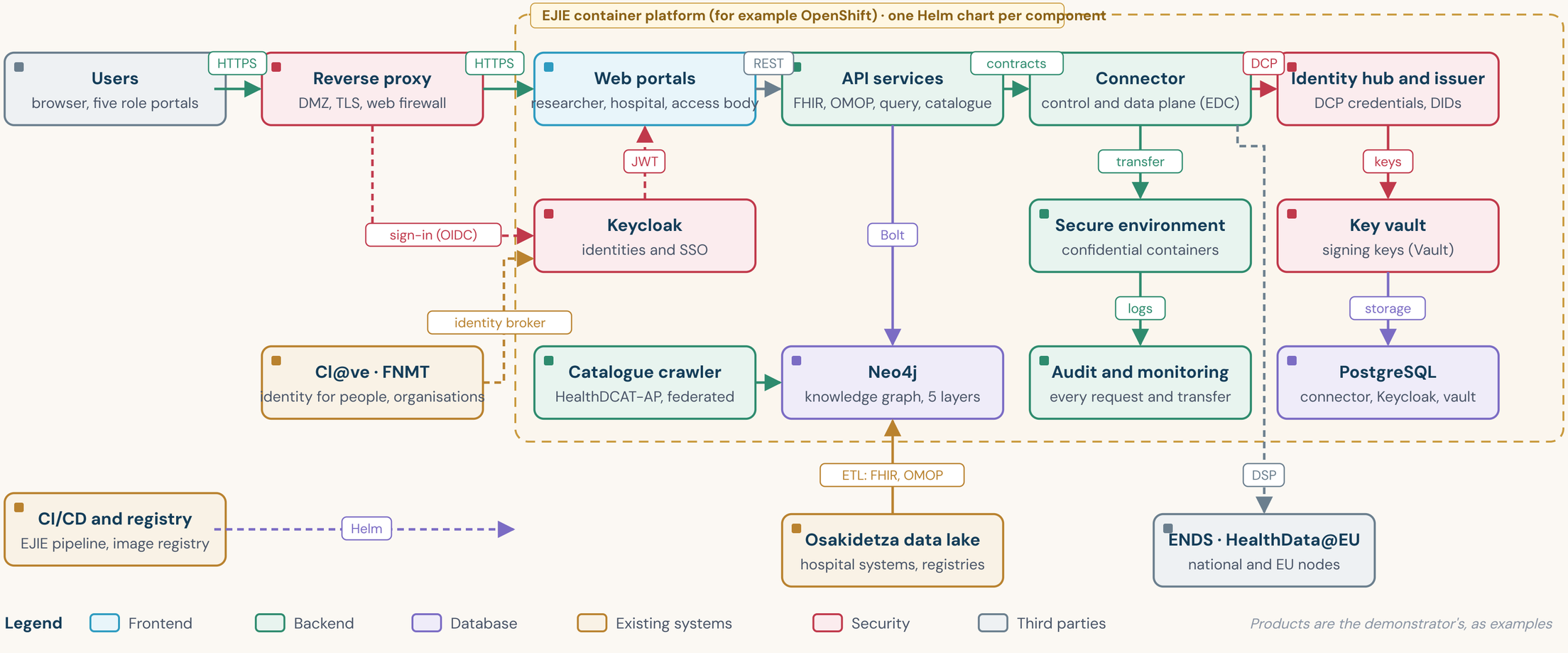


Every box exists in the demonstrator and runs today on Azure Container Apps. Grey lines name what implements it; for Euskadi these are examples, not a choice.

QUESTION 4 · ARCHITECTURE

High-level architecture: deployment at EJIE

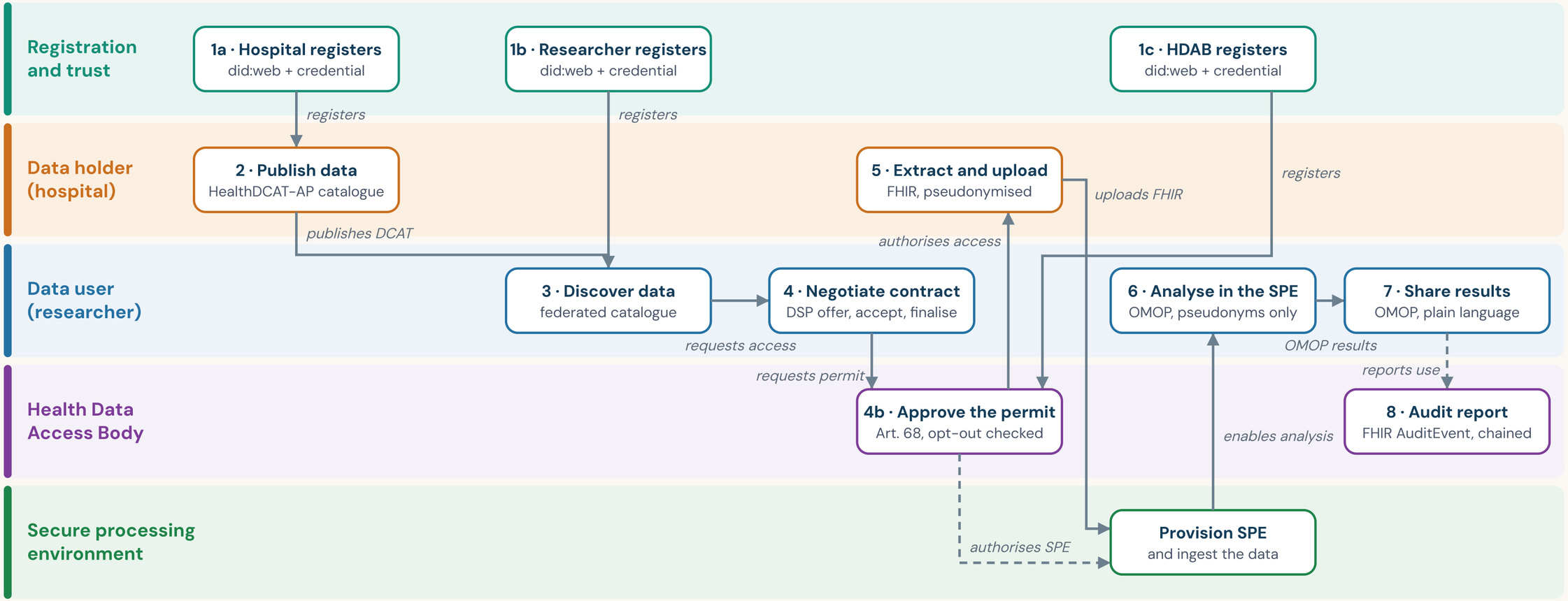
Every component runs in a container on EJIE's platform, one Helm chart each.



QUESTION 4 · ARCHITECTURE

The researcher's journey, step by step

From registration to an audited result: five roles, eight steps, all running in the demonstrator today.



QUESTION 5 · MATURITY

How mature are the components?

The components follow the service map. Health standards are mature; secure environments are not yet.

The service map

- Connectors
- Identity and credentials
- Catalogue
- Onboarding
- Conformity testing

Stable

Dataspace Protocol

On its way to an ISO standard, ISO/IEC 26450.

Mature

FHIR and OMOP

Health standards used around the world.

Changing

Dataspace connector, for example Eclipse EDC

In production since 2023 (Catena-X runs its Tractus-X distribution), still version 0.x. Breaking changes every quarter.

Young

Secure processing and data permits

The least mature parts. Plan extra time and budget here.

QUESTION 6 · GOVERNANCE

Three roles kept apart

Catena-X separates the rules, the code and the operation. We suggest the same here.

RULES · IN CATENA-X

Catena-X Association

Sets standards, runs certification.

CODE · IN CATENA-X

Eclipse Foundation

Hosts the open-source code.

OPERATION · IN CATENA-X

Cofinity-X

Runs the operation.

EUSKADI

Access body: open point

three options, two slides on.

EUSKADI

Open-source projects

used, and improved together.

EUSKADI

EJIE

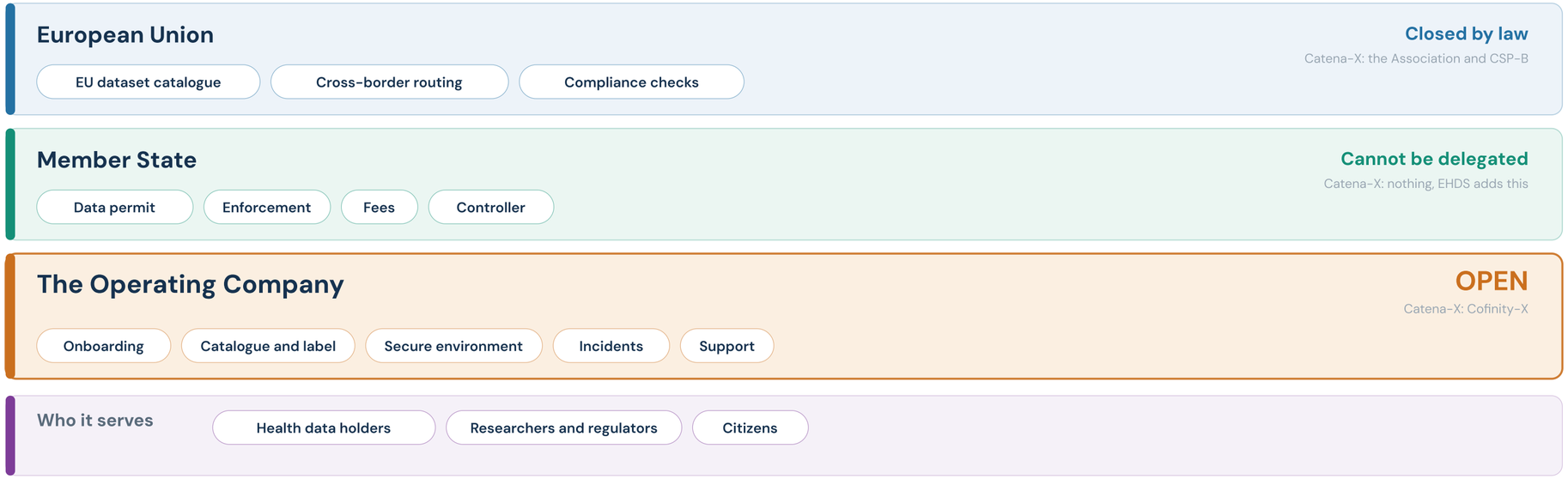
runs the core services.

Keep rules and operation apart. Design the governance for several domains from the start.

QUESTION 6 · GOVERNANCE

Who runs what: the Catena-X operating model

A service map with clear roles. The EHDS defines the roles, but not who runs them.



Regulation (EU) 2025/327, Chapter IV · the full model in docs/ehds-operating-company.md

Implemented: the Catena-X operating model, with a service map and clear roles.

Considered: the EHDS role model alone. It has no answer for operations and onboarding.

QUESTION 6 · GOVERNANCE

From Catena-X to a health data space

Who does what. The EHDS adds two functions Catena-X never had: data permits and the secure environment.

FUNCTION	IN CATENA-X	IN A BASQUE HEALTH DATA SPACE	BASIS
Rule-setting	The Association: standards and the rulebook	The Health Department with the access body; hospitals in the working groups	EHDS Art. 55
Data permits	none: contracts between companies	The access body (open point: which one)	Art. 68
Onboarding	The operator and onboarding service providers	EJIE as operator, with each hospital's IT	rulebook
Catalogue	The operator runs discovery	The access body owns it, EJIE runs it, in HealthDCAT-AP	Art. 77
Secure environment	none	The access body is responsible; EJIE or ENDS runs it (open point)	Art. 73
Certification	The Association, through conformity assessment bodies	Conformity tests for connectors; a quality and utility label for datasets	rulebook, EHDS
Incidents	The operator, with the participants	EJIE runs incident response; personal data breaches go to the data protection authority	ENS, GDPR Art. 33
Support	The operator and its service providers	EJIE service desk; each hospital for its own systems	service levels

QUESTION 6 · GOVERNANCE · OPEN POINT

Open point: who is the access body?

The Basque Health Data Office was announced in 2025. Its role under the EHDS is not decided yet.

A

A Basque access body

The Basque office is an access body, with a national coordinator (Art. 55(1)).

For: decisions close to the data.

Against: the Ley de Salud Digital must allow it.

B

One national access body

Spain designates one body. The Basque office prepares requests and runs the regional node.

For: one door for applicants.

Against: less regional control.

C

Start now, decide later

The pilot starts under a cooperation agreement. The role is settled when the law passes.

For: the pilot is not blocked.

Against: the role must be revisited.

Spain designates its access bodies by 26 March 2027. The pilot can start under any of the three.

Regulation (EU) 2025/327, Art. 55(1) · Ley de Salud Digital, draft of September 2025

QUESTION 7 · COSTS

No licences, but people and hosting

The core stack is open source. The money goes into engineering, hosting and operations.

EUSKADI REGIONAL PILOT, OUR ESTIMATE · PENDING FURTHER DETAIL FROM EJIE

€0.8 – 1.5M

To build the pilot

Engineering, mapping and onboarding. Assumptions on the next slide.

15 – 25% a year

To operate it

Of the build cost: about €120,000 – 375,000 a year.

€3,000 a month

Hosting: prod, test and dev

The monthly price depends on the hyperscaler. Plus SDLC and developer tools; the secure environment comes on top.

QUESTION 7 · COSTS · ASSUMPTIONS

The assumptions behind the estimate

What €0.8 to 1.5M includes, and what it does not. To be refined with EJIE's detail.

Duration and rate

12 months to build, with 8 to 15 full-time people. That is about €100,000 per person-year, or roughly €450 per person-day, blended across the profiles.

Scope

2 to 3 Osakidetza hospitals and the hospital cancer registry. Two use cases, one regional node. Mapping the data to FHIR and OMOP for those use cases is included.

Not included

Hospital staff time (clinical coders, IT). The data lake, already funded. The secure processing environment. The access body's own staff. Tender costs and VAT.

Hosting, about €3,000* a month

Production, test and development: connector, catalogue, identity and monitoring on managed containers. Priced by the hyperscaler; not the secure environment.

The secure environment is its own line, sized with the use cases: several thousand euros a month.

For scale: in our demonstrator one confidential compute node costs about 150 to 300 US dollars a month while it runs. A pilot adds research workspaces, storage, backup and ENS controls.

* This cost depends on the cloud provider. · EJIE Data Space · Euskadi

QUESTION 8 · TEAM

8 to 15 people for a pilot

Full-time people, with the right mix of skills.

8–15

full-time people

The experts need time and a mandate from top management.

PYRAMID TO BE SIZED ACCORDING TO THE PROJECT DETAILS

PROFILE TYPOLOGY · PEOPLE PER PROFILE

1–2	**Architect**	Holds the whole picture.
2–3	**Connector engineers**	Run and update the dataspace connectors.
2–3	**Platform and security (DevSecOps)**	Hosting, identity, monitoring and security.
2–5	**FHIR, OMOP and terminology**	Make the data mean the same everywhere. The largest task.
1–2	**Data protection and legal**	GDPR, EHDS, data permits and contracts.

THE SUGGESTED PILOT · A REGIONAL EHDS NODE

Euskadi pilot: start regional, connect upwards

Build one Basque node first. Then connect it to Spain, and through Spain to the EU.

connect upwards

connect upwards

STEP 1 · 2027–28 · BUILD

Euskadi pilot

- Basque Health Data Office
- Data lake → FHIR and OMOP
- One regional node
- Secure processing environment
- Two use cases

STEP 2 · 2028 · ALIGN

Connect to Spain

- ENDS, the national data space
- National access body
- Ley de Salud Digital
- MyHealth@EU (primary use)

STEP 3 · MAR 2029 · REUSE

Connect to the EU

- HealthData@EU
- EHDS implementing acts (2027)
- EUDI Wallet and eIDAS 2

FOUNDATION · AVAILABLE TODAY · REUSE FROM DAY ONE

Eclipse EDC · Simpl-Open · HealthData@EU code · FHIR · OMOP · SNOMED CT · Cl@ve · FNMT · ENS

available · in progress · open decision, clarify with the Ministry · we build it

THE PILOT · STEP 1 OF 3 · 2027–2028

Step 1: build the Euskadi pilot

Build with Osakidetza and EJIE. Who acts as access body is still open (question 6).

Basque Health Data Office: announced, its EHDS role still open
Announced in January 2025. A model: Aragón opened Spain's first regional office, October 2025.
in progress

Osakidetza data lake → FHIR and OMOP
The data lake is Euskadi's part of the national ENDS: €1.35M of EU recovery funds, February 2025.
in progress

One regional node: connector and metadata
A dataspace connector (for example Eclipse EDC) and a HealthDCAT-AP catalogue, operated by EJIE.
we build it

Secure processing environment: our own, or national?
A Basque environment, or the processing capacity of ENDS. Either way it must meet the ENS.
open decision

Two use cases: cohort discovery and a disease registry
For example the hospital cancer registries that Osakidetza already runs.
we build it

THE PILOT · STEP 2 OF 3 · 2028

Step 2: connect to Spain

Align with the Ministry of Health and the Ministry for Digital Transformation (MTDFP).

ENDS, the National Health Data Space: the interface for regional nodes

Opened January 2026: a network of regional platforms on common protocols, with €28M for the regions. ENDS home

in progress

The access body for Spain: one national, or regional ones with a coordinator?

It must be designated by 26 March 2027. The Regulation allows several bodies with one coordinator: Art. 55(1).

open decision

Ley de Salud Digital: the legal role of the regions

Still a draft (public consultation from September 2025). It sets up the access bodies and both national and regional catalogues.

open decision

MyHealth@EU national contact point: HCDSNS and e-Receta

Primary use, and already live: Euskadi is connected to the EU service since 2023. Proof that the path upwards works. MyHealth@EU

available

THE PILOT · STEP 3 OF 3 · MARCH 2029

Step 3: connect to the EU, on a ready foundation

Reuse what the European Commission builds. Most of the foundation is available today.

HealthData@EU: cross-border access for research
Connection due 26 March 2029, and 2031 for the remaining data categories. EHDS
in progress

EHDS implementing acts: catalogue, permits, quality label
Due earlier, by 26 March 2027: they shape what Step 1 builds. Regulation (EU) 2025/327
in progress

EUDI Wallet and eIDAS 2: trust for citizens and users
Every Member State must offer a wallet by the end of 2026. European Digital Identity
in progress

FOUNDATION · AVAILABLE TODAY · REUSE FROM DAY ONE

Eclipse EDC · Simpl-Open
Open-source connector; Simpl-Open release of July 2026.

HealthData@EU platform code
Open source on code.europa.eu since May 2025.

FHIR · OMOP · SNOMED CT
FHIR R4, OMOP CDM and SNOMED CT: mature, used worldwide.

Cl@ve · FNMT · ENS (CCN)
Cl@ve identity, FNMT certificates, the ENS security scheme.

WHAT COULD COME NEXT

From a first definition to an operational pilot

Five phases. Each ends with a deliverable and a clear criterion for moving on.

PHASE	ACTIVITIES	OWNER	DURATION	DELIVERABLE	MOVE ON WHEN
1 Define	Choose one or two use cases, agree scope and roles, name a sponsor	EJIE, with the Health Department and Osakidetza	2–3 months	Pilot charter	Use cases and sponsor approved
2 Prepare	Working groups with the hospitals, data inventory, target architecture, tender on standards and conformity tests, impact assessment	EJIE, with the hospitals	3–4 months	Tender specification	Budget committed, specification approved
3 Tender	Publish, evaluate, award	EJIE procurement	6–9 months	Signed contract	Contract awarded
4 Build	Node, identity, catalogue, mapping; onboard 2 to 3 hospitals	Supplier, with EJIE	9–12 months	A node that passes the conformity tests	Tests passed, ENS accreditation, impact assessment updated
5 Operate	First data permits, measure the use cases	EJIE; the access body for permits	6 months	Pilot report with measured results	Decision to scale and connect to ENDS

March 2027

Access bodies designated, during phase 2.

March 2029

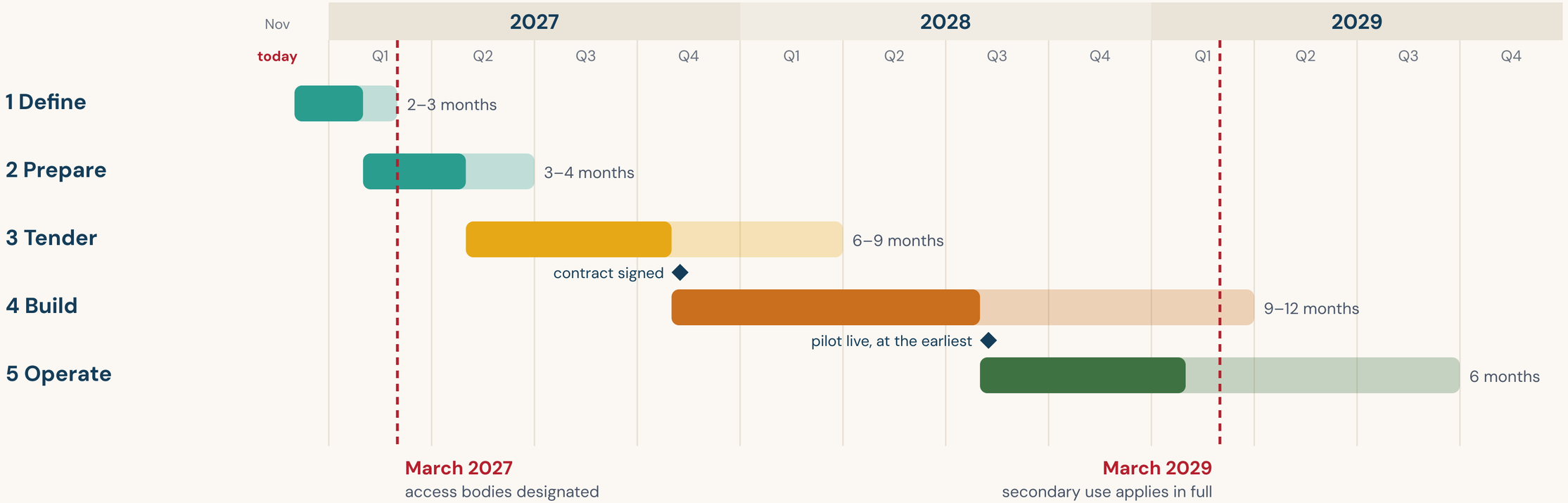
Secondary use applies in full. Starting phase 1 now, the pilot is live around then.

Regulation (EU) 2025/327, Art. 55(6) and 105 · durations are indicative

WHAT COULD COME NEXT · TIMELINE

The calendar: from today through March 2029

If the definition starts now, the pilot is live between autumn 2028 and spring 2029. The tender sets the pace.



Dark: the shortest duration from the earliest start. Light: how far the phase can run with the longest durations. Durations are indicative.

LET'S DISCUSS

Three questions to start our discussion

1. What are the first two use cases that would demonstrate value?
2. Who will set the rules, and who will operate the services?
3. Which hospitals would join the first working group?

Eskerrik asko · Gracias · Thank you

Live demonstration: ehds.mabu.red

Matthias Buchhorn

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